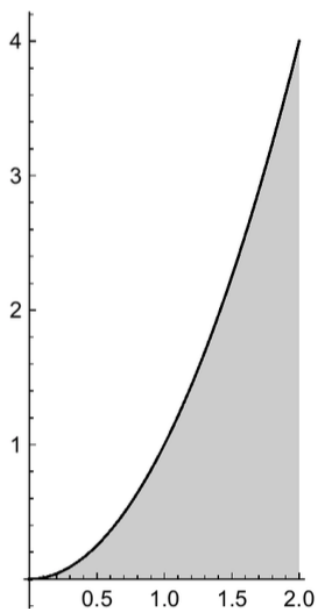


Problem 5

Problem 5

Problem 1 The graph of the function $f(x) = x^2$ is given in the figure.



Find the exact value of the area of the shaded region. *HINT: Use a right Riemann sum with n rectangles, and then take the limit as $n \rightarrow \infty$.*

Explanation. $\Delta x = \frac{2}{n}$
 $x_k^* = x_k = k\Delta x = k\left(\frac{2}{n}\right) = \frac{2k}{n}$, for $k = 1, \dots, n$

$$A \approx \sum_{k=1}^n f(x_k^*)\Delta x = \Delta x \cdot \sum_{k=1}^n f(x_k^*) = \frac{2}{n} \cdot \sum_{k=1}^n f\left(\frac{2k}{n}\right) = \frac{2}{n} \cdot \sum_{k=1}^n \left(\frac{2k}{n}\right)^2 =$$

$$\frac{2}{n} \cdot \sum_{k=1}^n \frac{4k^2}{n^2} = \frac{8}{n^3} \cdot \sum_{k=1}^n k^2$$

Recall: $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$

Therefore

Problem 5

$$A \approx \frac{8}{n^3} \cdot \sum_{k=1}^n k^2 = \frac{8}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{4}{3} \cdot \frac{(n+1)(2n+1)}{n^2} = \frac{4}{3} \cdot \frac{2n^2 + 3n + 1}{n^2}.$$

Let's take the limit of Riemann sums as $n \rightarrow \infty$.

$$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x = \lim_{n \rightarrow \infty} \frac{4}{3} \cdot \frac{2n^2 + 3n + 1}{n^2} = \lim_{n \rightarrow \infty} \frac{4}{3} \cdot \left(2 + \frac{3}{n} + \frac{1}{n^2} \right) = \frac{8}{3}.$$